

Sarkar Marshia Hossain

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CAREER HIGHLIGHTS:

- 2+ years of professional work experience spanning academic research, teaching, and academic labs.
- Expertise in MatLab & AutoCAD.
- Self-driven & fast learner with strong problem-solving, and analytical skills; excel in multi-disciplinary environments.
- Lawful US permanent resident (Category: E26), eliminating the need for visa sponsorship.

PROFESSIONAL EXPERIENCE:

The University of Toledo, OH, USA

Graduate Teaching Assistant, Electrical Engineering & Computer Science Department **Aug 2021 – May 2023**

- Designed a Communication model between Satellite and Connected vehicles using LDPC coding and QAM modulation for Digital Video Broadcasting.
- Mentored undergraduate-level electronics based project.

Robi Axiata Limited, Dhaka, Bangladesh

Power System Engineering Intern

May 2019 – August 2019

- Designed the process of the solar power system.
- Explored Base Station System (BSS) and became acquainted with the operation of electronics devices.

TECHNICAL SKILL:

- **Certification:** Basic Python, Data Structure with Python, Cisco CCNA.
- **Vulnerability Management:** Vulnerability scanning, detection, risk analysis, risk prioritization, and mitigation
- **Design Software:** AutoCAD, VLSI, CST studio, Microwind, PSpice, HFSS, Dsch2.
- **Simulator:** Cisco Packet Tracer, Simulink.
- **Programming:** MATLAB, Python, C.
- **Soft Skills:** Time Management, Multi-tasking, Teamwork, Problem-solving, Documentation, Presentation, Communication, and Critical Thinking.

EDUCATION:

M. Sc. in Electrical Engineering | The University of Toledo, OH, US

Aug 2021 - August 2023

- **Thesis Topic:** Performance of LDPC Code for Video Transmission Between Satellites and Connected Vehicles in Presence of AWGN Channel.
- **Relevant Coursework:** Signal and System, Digital Signal Processing, Power System Management.
- **Relevant topics:** Power Generation, Distribution, Transmission, Critical Mathematical Problems solution.
- **CGPA:** 3.95 out of 4.00.

B.Sc. in Electronics & Telecommunication Engineering | Rajshahi University of Engineering & Technology, Rajshahi, Bangladesh

Jan 2015 – Oct 2019

- **Relevant Coursework:** Control System, Power Electronics, Fiber Optics Communication, Antennas and Propagation, Satellite and Solar System.
- **Relevant Topics:** Digital and Electronics Circuits, Motor, Generator, 3-Phase electrical Circuit.
- **CGPA:** 3.64 out of 4.00

Academic Projects

Design of fire sensing circuit

- Tools Utilized: Transistor, Sensor, transmitter, Buzzer, Microcontroller.
- Designed an electronics circuit using a BC457 transistor to sense fire.

Calculation of reliability metrics budget and capital cost of a turbine gas plant

- Researched and Analyzed of SAIDI, SAIFI, CAIDI, MAIFI and capital cost of the gas plant.

SELECTED PUBLICATIONS: ([Google Scholar Link](#))

- **S. M. Hossain**, M. Y. Ali, M. S. Hossain and M. F. Reza, "Performance Investigation of Robust Cylindrical Antenna Array Beamformer in Appearance of Look Direction Mismatch," 2019 4th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2019, pp. 1-5, doi: 10.1109/EICT48899.2019.9068803.
- **S. M. Hossain**, M. Y. Ali, M. S. Hossain and M. F. Reza, "Robust Cylindrical Antenna Array using Optimal Diagonal Loading Technique in the Presence of Steering Angle Mismatch," 2019 3rd International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE), Rajshahi, Bangladesh, 2019, pp. 276-279, doi: 10.1109/ICECTE48615.2019.9303584.
- **S. M. Hossain**, Alam, S.M., Oluoch, J. and Kim, J., 2022, October. A Novel Secure Physical Layer Key Generation Method in Connected and Autonomous Vehicles (CAVs). In 2022 IEEE Conference on Communications and Network Security (CNS) (pp. 1-6). IEEE.

LEADERSHIP:

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| • Operations Assistant, UT Recreation Center, Toledo, OH | May 2022-August 2022 |
| • General Secretary, Bangladesh Student Association (BSA), University of Toledo | July 2022 – June 2023 |
| • Assistant General Secretary of RUET Career Forum (RCF) | Jan 2019 - Dec 2019 |
| • Vice President, Pabna District Association of RUET | Jan 2018 – Dec 2018 |